**RestNet**

UNDERSTAND YOUR SLEEP

YA **User Y.A.**
9 – 22 February 2026

NAVIGATION

- Sleep Dashboard
- Night Explorer
- My Sleep Insights

Environmental Sensing for Sleep Quality Analysis

IoT & Applications
Imperial College London
March 2026

# Night Explorer

Dive Into a Single Night

Select a night below to explore what happened while you slept. This page breaks down your **sleep architecture**, shows a minute-by-minute **bedroom environment timeline** with a **quick summary** of the night, checks whether **outdoor air quality** met WHO guidelines, and explores **what was different** about this night.

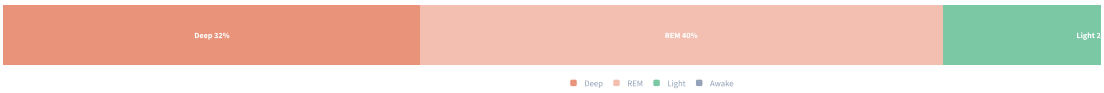
Please Select a Night

2026/02/22

## Sleep Architecture

| SLEEP SCORE | VS. 2-WEEK AVERAGE SCORE | TOTAL SLEEP TIME |
|---|---|---|
| 87 / 100 | +4 points | 9 h 28 m |

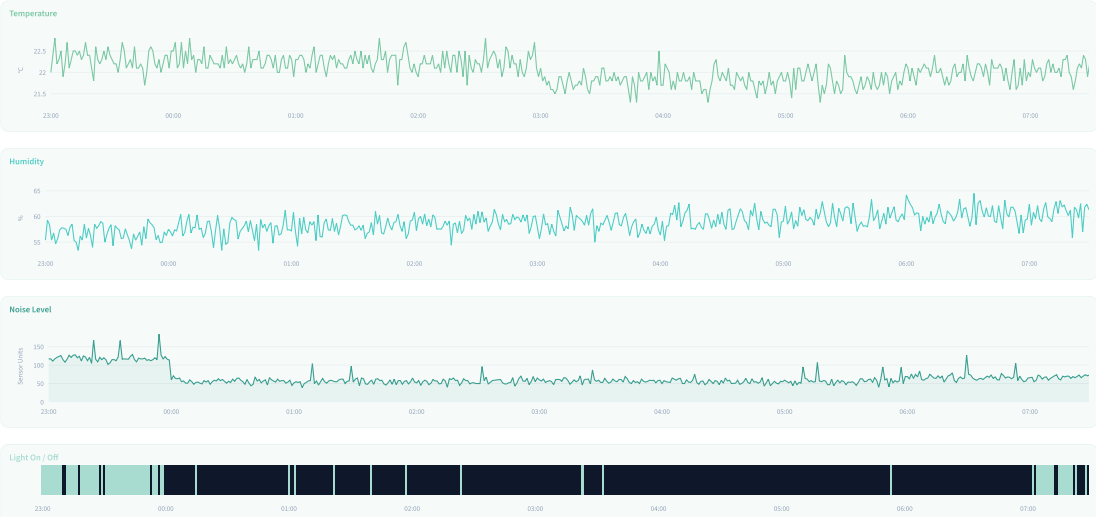
Sleep Stages Breakdown




## Bedroom Environment Timeline

Minute-by-minute bedroom conditions from 11 pm to 9 am. Hover for exact values.

These are the four bedroom conditions recorded by your sensors throughout the night. Temperature, humidity, and noise level were all linked to sleep quality across your 14 nights. Light exposure is included for completeness but did not show a significant link to sleep quality.




## Night at a Glance

A quick snapshot of your bedroom conditions for this night.

Temperature, humidity, and noise were associated with sleep quality across your 14 nights. Light exposure is included for general context.

### Highs & Lows

| | | | |
|---|---|---|---|
| **21.8°C at 3 AM** | **22.3°C at 11 PM** | **4 AM, avg 55** | **11 PM, avg 120** |
| Coolest hour | Warmest hour | Quietest hour | Noisiest hour |

### Averages & Comparison

| | | |
|---|---|---|
| **59.2%** | **67** | **153 of 600 min (26%)** |
| Average humidity (14-night avg: 50.8%) | Average noise level (14-night avg: 78) | Light exposure |

## Air Quality Verdict

✓ **10.4** $\mu g/m^3$
Overnight PM2.5 averaged 10.4 $\mu g/m^3$, within WHO 24-hour guideline (15)

✓ **18.2** $\mu g/m^3$
Overnight $NO_2$ averaged 18.2 $\mu g/m^3$, within WHO 24-hour guideline (25)

*Both pollutants stayed within WHO guidelines overnight. PM2.5 peaked at 14.7 $\mu g/m^3$ and $NO_2$ at 31.7 $\mu g/m^3$.*

*Neither PM2.5 nor $NO_2$ showed a significant link to sleep quality in this dataset. These are included for general health context against WHO guidelines. WHO guidelines for PM2.5 (15 $\mu g/m^3$) and $NO_2$ (25 $\mu g/m^3$) are based on 24-hour averages. This comparison uses overnight data only (approx. 11 pm to 9 am), so values shown represent a partial-day average rather than a full 24-hour measurement.*

## What Was Different?

How this night's conditions compare to your usual, focusing on factors most associated with your sleep quality.

- Better than usual
- Close to usual
- Worse than usual

**Noise variability:** 20.8 (your median: 21.0)
*Close to your usual level. Noise variability mattered more than any other bedroom condition.*

**Noise level:** 66.9 (your median: 75.8)
*A quieter night, linked to better sleep.*

**Humidity:** 59.2% (your median: 50.4%)
*Higher humidity linked to lower sleep scores.*

**Temperature stability:** 1.5°C (your median: 1.5°C)
*Temperature range was typical for you.*

*Light exposure, PM2.5, $NO_2$ showed no statistically significant correlation with sleep quality across your 14 nights of data.*